

Lab 3 — Architectural Justification

System: Academic Elective Bidding & Allocation System | **Domain:** Campus & Academic Operations | **Stakeholders:** Student, Academic Registrar

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Architecture Selection: We chose Layered Architecture for the Academic Elective Bidding & Allocation System.

Architectural Choice

The system is organised into a Presentation Layer (student bidding portal and registrar admin console), a Business/Service Layer (Bid Manager, Allocation Engine and Notification Service components) and a Data & External Layer (Database component). Requests flow strictly downward through the layers, with the Bid Manager Component acting as the central coordinator between the UI, the allocation logic, notifications and persistent storage.

Two Reasons

1. **Predictable, single-campus workload:** elective bidding runs within a fixed bidding window for one institution's student body, not at continuous, unbounded scale. Layered architecture gives clear separation of concerns (UI, business rules, data) without the operational overhead — service orchestration, distributed data consistency — that microservices would add for no real benefit here.
2. **Shared, centralised business rules:** both students and the Academic Registrar depend on the same bid-validation, seat-capacity and eligibility rules inside the Bid Manager Component. Keeping this logic in one layer avoids duplicating rules across independent services, which would risk the registrar's view of seat allocation drifting out of sync with what students see.

Security Advantage

Because the Presentation Layer never talks to the Database Component directly, all student bid data and registrar actions are validated and role-checked (student vs. registrar privileges) inside the Business Layer before any data access occurs. This layer boundary is a natural enforcement point for authentication and role-based access control, reducing the attack surface that would otherwise directly expose sensitive academic records.

Performance Benefit

Since the Allocation Engine, Bid Manager and Database components sit within one logical application boundary rather than communicating over a network as separate services, the allocation run — matching every submitted bid against elective seat capacity — executes with in-process calls instead of incurring inter-service network latency. This keeps the end-to-end allocation and confirmation well within the required processing window during peak bidding periods.

Lab 3 — Component Diagram

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